I- Personal Information:

Full Name (English)	Elham Fahad Alkhammash
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II- Research Interests:

Discipline:	Materials Science and their application
Subdiscipline:	Advanced glass and glass-ceramic materials for optical, photonic, and biomedical applications, with expertise in theoretical biophysics and computational modeling of biochemical and neural systems.
Research Interests	**Advanced glass and glass-ceramic materials for optical and photonic applications.** **Design, composition, synthesis, and processing of functional glass systems.** **Rare-earth-doped glasses: optical, spectroscopic, thermal, and structural characterization.** **Luminescent materials, photonic technologies, and optical sensing.** **Fiber-optic sensing and environmental optical applications.** **Theoretical biophysics and computational modeling of biochemical and neural systems.**

Research Projects

1. **Master's Thesis Research Project**, University of Ottawa, Canada (2020)
Modeling the Reaction Transport of Synaptic Acetylcholine Released at High Frequency in Weakly Electric Fish.
The research focused on reaction–diffusion modeling of acetylcholine transport,

nonlinear biochemical reactions, and synaptic transmission dynamics at high firing frequencies.

2. **PhD Research Project**, King Khalid University, Saudi Arabia (Ongoing)
   Thermal Stability and Luminescence Behavior of Rare-Earth-Doped TeO_2–ZnO–Nb_2O_5–KF–La_2O_3 Tellurite Glasses for Photonic Applications.
   The research focuses on the development and characterization of rare-earth-doped tellurite glasses for photonic and optical applications, including thermal, structural, optical, and luminescence properties.

III- Published paper :

Alkhammash, E., L'Heureux, I., Morris, C. E., & Joós, B. (2021). Reaction–diffusion modelling to assess what limits effective acetylcholine fluctuations at high-frequency cholinergic electroplaques. Biophysical Journal.
https://www.cell.com/biophysj/fulltext/S0006-3495(22)03782-1

Alkhammash, E., L'Heureux, I., Morris, C. E., & Joós, B. (2023). Using reaction–diffusion modeling to probe high-frequency cholinergic transmission. Biophysical Journal.

https://www.researchgate.net/profile/Elham-Alkhammash

under review.

Alkhammash, E., Sultan, B., Ibrahim, A., & Yousef, E. Impact of physical parameters and elasticity of composition: TeO_2–ZnO–Nb_2O_5–KF–La_2O_3. under review.

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